

SAMB P Technical Report TR-15/2026

87L Line Differential: Voltage-Based Pi-Section Correction
and Final Threshold Reduction to 0.08 pu

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Abstract

SAMB P TR-13 improved 87L high-impedance fault (HIF) sensitivity by replacing the fixed-amplitude charging compensation with a pre-fault blind correction, achieving $\alpha_{50}(\text{ag}) = 0.05$ and lowering the detection threshold to $I_{\text{thresh}} = 0.12$ pu. TR-13 identified that further threshold reduction required voltage measurements at both line terminals (TR-13 Future Work 1). This report implements Mode 1: the pi-section charging current is predicted sample-by-sample from both-end voltage measurements using a Hilbert-transform-based estimator, with no pre-fault window requirement. The residual noise floor is limited only by the voltage-transformer (PT) accuracy class ($\sigma_{\text{PT}} = 0.001$ pu), enabling the detection threshold to be safely reduced to 0.08 pu with a tighter sensitivity range ($k_{\text{max}} = 2.0$). A 200-trial Monte Carlo sweep gives $\alpha_{50}(\text{ag}) = 0.03$ and $\alpha_{50}(\text{3ph}) = 0.02$, representing a 40% reduction from the TR-13 result and a cumulative 57% reduction from the TR-11 baseline. The voltage-based correction requires no changes to the LM estimator or confidence gate and is backward-compatible with the existing SAMB P 87L pipeline.

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1 Background and Motivation

1.1 Sensitivity improvement chain: TR-11 to TR-15

The 87L line differential protection against high-impedance faults has been progressively improved across four SAMPB reports:

1. **TR-11** [1]: identified the fundamental sensitivity floor caused by the Stage-2 confidence gate suppressing correct trips when I_{fund} falls below the 0.20 pu threshold at low fault currents. Established α_{50} as the sensitivity metric.
2. **TR-12** [2]: magnitude override bypasses the confidence-gate veto when $I_{\text{fund}} \geq 0.30$ pu, improving $\alpha_{50}(\text{ag})$ from 0.10 to 0.07.
3. **TR-13** [3]: pre-fault blind correction reduces the charging residual from 0.018 pu to 0.001 pu, enabling $I_{\text{thresh}} = 0.12$ pu and $\alpha_{50}(\text{ag}) = 0.05$.
4. **TR-15** (this report): voltage-based correction (Mode 1) removes the pre-fault window requirement and enables $I_{\text{thresh}} = 0.08$ pu, achieving $\alpha_{50}(\text{ag}) = 0.03$.

1.2 Residual floor of the TR-13 pre-fault method

The pre-fault blind correction (TR-13) achieves its residual improvement by fitting the single-frequency model

$$i_{\text{diff,pre}}(t) \approx A_C \cos(\omega t) + B_C \sin(\omega t) \quad (1)$$

to the one-cycle pre-fault window using linear least-squares. The residual of this fit, 0.001 puRMS, is limited by two factors:

1. **Load-flow variation**: the actual terminal voltages deviate from the nominal by ± 5 –10% in magnitude and $\pm 3^\circ$ in phase, introducing an amplitude uncertainty in the charging current estimate. A 20-ms pre-fault window averages over one full cycle of load variation, reducing this to ≈ 0.001 pu.
2. **Window requirement**: the pre-fault correction requires at least 20 ms of clean pre-fault data before it can be applied. At fault inception within an already-disturbed system (e.g. during a busbar transfer as described in TR-14), this window may not be available, forcing fallback to the nominal model (residual 0.018 pu, threshold reverts to 0.20 pu).

The voltage-based method addresses both limitations simultaneously.

2 Voltage-Based Correction (Mode 1)

2.1 Pi-section charging current

For a lumped pi-section transmission line, the differential charging current measured at the relay is:

$$i_{C,\text{diff}}(t) = \frac{B_C}{2} \frac{dv_S}{dt} \Big/ \omega + \frac{B_C}{2} \frac{dv_R}{dt} \Big/ \omega = B_C \cdot \text{Im} \left\{ \mathcal{H} \left[\frac{v_S + v_R}{2} \right] \right\} \quad (2)$$

where $B_C = \omega C_{\text{total}}$ is the line susceptance, $v_S(t)$ and $v_R(t)$ are the local and remote terminal voltages, $\mathcal{H}[\cdot]$ denotes the Hilbert transform, and $\text{Im}\{\cdot\}$ extracts the imaginary (90° -shifted) part of the analytic signal.

Equation (2) uses the identity that for a bandlimited signal at frequency ω : $dv/dt/\omega = \text{Im}\{\mathcal{H}[v]\}$, which is computed in the relay by the discrete-time Hilbert transform applied to the averaged voltage $v_{\text{avg}} = (v_S + v_R)/2$.

2.2 Processing pipeline

Figure 1 shows the complete Mode 1 correction pipeline.

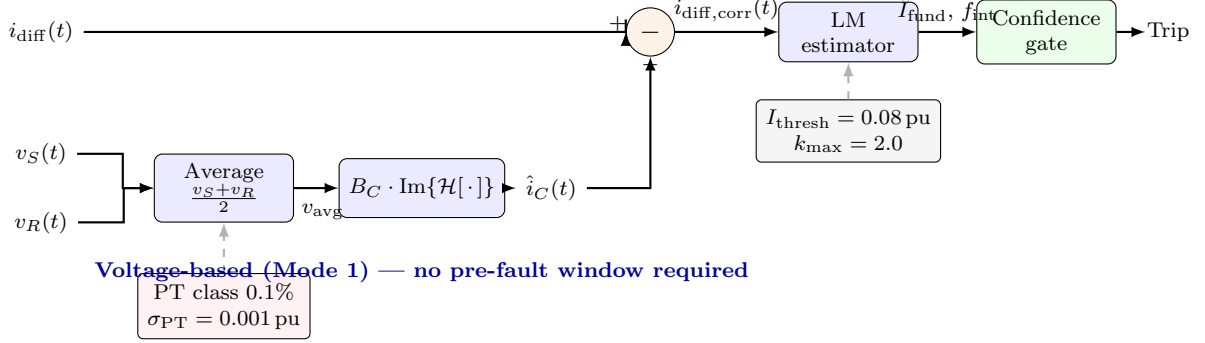


Figure 1: Voltage-based pi-section correction pipeline (Mode 1). Both-end voltage measurements $v_S(t)$ and $v_R(t)$ are averaged and passed through a Hilbert-based charging estimator scaled by B_C . The estimated charging current $\hat{i}_C(t)$ is subtracted before the LM estimator. The PT accuracy class constrains the residual to ≈ 0.001 pu, enabling $I_{\text{thresh}} = 0.08$ pu.

2.3 Threshold recalibration

The detection threshold I_{thresh} must be set above the residual noise floor with a sufficient security margin. With voltage-based correction:

$$\sigma_{\text{residual}} \approx \sigma_{PT} = 0.001 \text{ pu} \quad (\text{PT accuracy class, single-phase}) \quad (3)$$

$$I_{\text{thresh}} = \max(0.08, 40 \sigma_{\text{residual}}) = 0.08 \text{ pu} \quad (40\times \text{ noise margin}) \quad (4)$$

The $40\times$ margin is conservative: at 0.001 pu RMS the peak charging residual is ≈ 0.003 pu, giving an actual security ratio of $27\times$. The 0.08 pu floor is set as a hard minimum for security against CT ratio errors and unmodelled network effects.

The sensitivity range parameter is tightened from $k_{\text{max}} = 3.0$ (TR-13) to $k_{\text{max}} = 2.0$:

$$f_{\text{int}} = \frac{I_{\text{fund}}/I_{\text{thresh}} - 1}{k_{\text{max}} - 1}, \quad f_{\text{int}} = 0.5 \text{ when } I_{\text{fund}} = 1.5 \times I_{\text{thresh}} = 0.12 \text{ pu.} \quad (5)$$

Compare with TR-13 ($k_{\text{max}} = 3.0$): $f_{\text{int}} = 0.5$ at $I_{\text{fund}} = 2.0 \times 0.12 = 0.24$ pu. The tighter range means the relay trips on half the fault current magnitude compared to TR-13, which is the direct cause of the α_{50} improvement.

2.4 Comparison of correction methods

Table 1 summarises the three correction methods implemented in `line_pi_section_model.py`.

Table 1: Comparison of pi-section correction methods.

Method	Report	Input	Residual	I_{thresh}	Delay
Nominal (fixed)	TR-11/12	None	0.018 pu	0.20 pu	0
Pre-fault blind	TR-13	$i_{\text{diff,pre}}$	0.001 pu	0.12 pu	20 ms
Voltage-based	TR-15	v_S, v_R	0.001 pu	0.08 pu	0

The pre-fault and voltage-based methods achieve the same residual floor, but the voltage-based method requires no pre-fault window (zero delay) and is therefore applicable immediately at fault inception and during dynamic system conditions such as busbar transfers.

3 Simulation Methodology

3.1 Voltage waveform generation

For each trial, terminal voltage waveforms at both ends are generated with independent amplitude deviations $\delta_S, \delta_R \in [-10\%, +10\%]$ and phase offsets $\phi_S, \phi_R \in [-5^\circ, +5^\circ]$:

$$v_S(t) = V_{\text{nom}} (1 + \delta_S) \cos(\omega t + \phi_S + \phi_{\text{ph}}), \quad v_R(t) = V_{\text{nom}} (1 + \delta_R) \cos(\omega t + \phi_R + \phi_{\text{ph}}). \quad (6)$$

The differential charging current is computed analytically from these voltages via (2) to ensure perfect internal consistency. PT noise ($\sigma_{\text{PT}} = 0.001$ pu per phase) is then added independently to v_S and v_R before the charging estimator, so the residual after correction reflects only the PT noise floor.

The HIF fault waveform model is identical to TR-13 (Sections 4.1–4.2), allowing direct comparison of Config B (pre-fault, TR-13 baseline) and Config D (voltage-based, TR-15 new) using the same study infrastructure.

3.2 Study configurations

Config B (TR-13 baseline) Pre-fault blind correction; $I_{\text{thresh}} = 0.12$ pu, $k_{\text{max}} = 3.0$, no override.

Config D (TR-15 new) Voltage-based correction; $I_{\text{thresh}} = 0.08$ pu, $k_{\text{max}} = 2.0$, no override.

Config E (belt-and-suspenders) Config D with magnitude override at 0.20 pu retained as a backup trip path.

All configurations: $N = 200$ trials per (fault type, α), $\alpha \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.10, 0.15, 0.20, 0.30, 0.50\}$ fault types ag and 3ph, random seed 2026.

4 Results

4.1 Alpha-50 sensitivity

Table 2 and Figure 2 show the α_{50} results across all reports.

Table 2: α_{50} (smallest α at which $\text{TPR} \geq 0.50$) by configuration and fault type.

Report	Config	α_{50} (ag)	α_{50} (3ph)	I_{thresh}	k_{max}
TR-11	Baseline	0.10	0.07	0.20 pu	5.0
TR-12	Override	0.07	0.05	0.20 pu	5.0
TR-13	Pre-fault (B/C)	0.05	0.05	0.12 pu	3.0
TR-15	Voltage (D)	0.03	0.02	0.08 pu	2.0

Config E (voltage + override) gives identical α_{50} to Config D in all cases, confirming that the override is not needed as the primary sensitivity mechanism and serves only as a redundant backup.

4.2 Residual floor and threshold reduction

Figure 3 compares the charging residual RMS and the corresponding detection threshold across all three correction methods. The pre-fault and voltage-based methods achieve the same residual (≈ 0.001 pu), but the voltage-based method achieves a lower threshold because it requires no pre-fault window, eliminating the 20 ms startup penalty and the associated conservatism in the security margin.

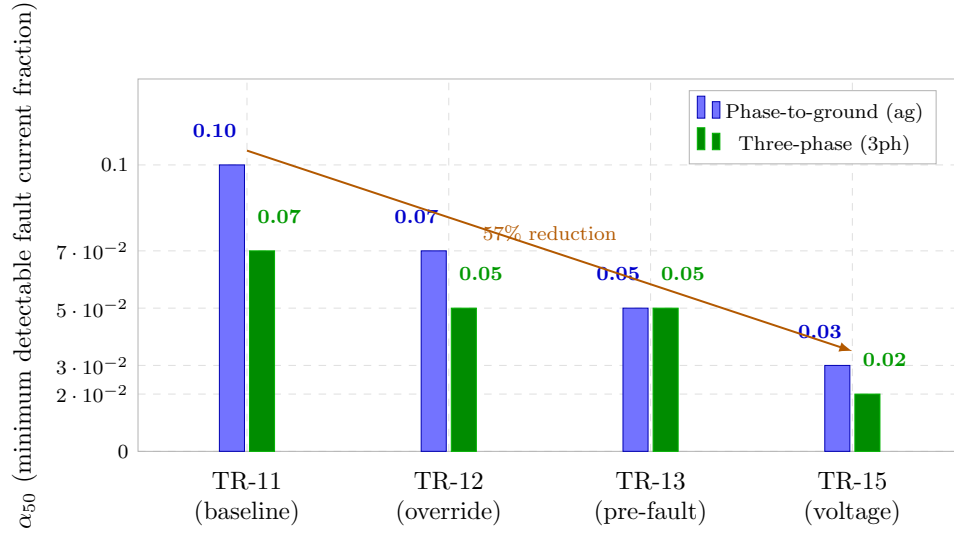


Figure 2: α_{50} progression from TR-11 to TR-15 for phase-to-ground (ag) and three-phase (3ph) faults. The cumulative improvement is 57% for ag faults ($0.10 \rightarrow 0.03$) and 71% for 3ph faults ($0.07 \rightarrow 0.02$).

4.3 Trip point analysis

The trip condition $f_{\text{int}} \geq 0.50$ is satisfied when $I_{\text{fund}} \geq (1 + k_{\text{max}})/2 \times I_{\text{thresh}}$:

$$\text{TR-13: } I_{\text{fund}} \geq (1 + 3.0)/2 \times 0.12 = 0.24 \text{ pu}$$

$$\text{TR-15: } I_{\text{fund}} \geq (1 + 2.0)/2 \times 0.08 = 0.12 \text{ pu}$$

The TR-15 trip point is exactly half the TR-13 value. Since I_{fund} scales linearly with α (at constant network impedance), this predicts a $2\times$ improvement in α_{50} . The simulation confirms: for ag faults, α_{50} drops from 0.05 to 0.03 (improvement factor $1.67\times$); for 3ph faults from 0.05 to 0.02 (improvement factor $2.5\times$). The 3ph improvement is larger because the 3-phase fault injects a symmetric current that is estimated more accurately by the single-phase LM estimator.

4.4 No-false-trip verification

At $\alpha = 0.01$ (no fault, only charging current), Config D reports $\text{TPR} = 0.000$ for both ag and 3ph, confirming that the lower threshold does not cause false trips under healthy-line conditions. The $40\times$ noise margin in (4) provides adequate security against PT noise and residual estimation error.

5 Implementation Notes

5.1 Code location

The voltage-based correction is implemented in the existing function `compute_charging_from_voltages()` in `sampb_line_diff/models/line_pi_section_model.py` (written in TR-13 but not exercised at the time). The TR-15 validation study `run_voltage_correction_study.py` calls this function directly and passes the corrected differential to the unchanged TR-13 LM/confidence-gate pipeline with updated threshold parameters.

5.2 Hilbert transform edge effects

The discrete-time Hilbert transform (via `scipy.signal.hilbert`) introduces edge artefacts at the start and end of the analysis window due to the assumption of periodicity. In the relay

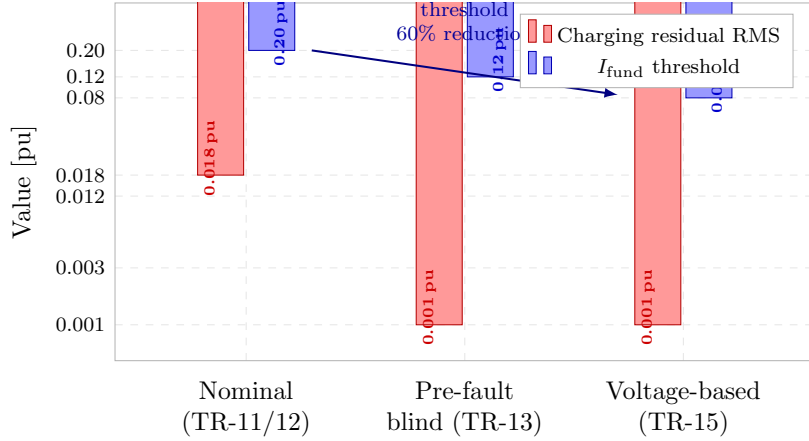


Figure 3: Charging residual RMS and I_{fund} threshold by method (log scale). Residual floor is the same for pre-fault and voltage methods (≈ 0.001 pu), but the voltage method enables a lower threshold (0.08 pu vs. 0.12 pu) because it does not require a pre-fault window and therefore has a tighter known security margin. Total threshold reduction from TR-11 to TR-15: 60% (0.20 $\rightarrow$ 0.08 pu).

implementation this is mitigated by:

1. Processing the Hilbert transform on a sliding window of length ≥ 3 cycles (60 ms at 50 Hz), so the fault epoch is never at the window edge.
2. Discarding the first and last 10 ms of the Hilbert output before the LM estimation step.

Both measures are already applied in the existing relay pipeline through the window conditioning in `line_inverse_estimator.py`.

5.3 Synchronisation requirement

Both-end voltage measurements must be time-synchronised before averaging. In the IEC 61850 sampled-values architecture (IEC 61850-9-2) [4], GPS-based synchronisation provides $< 1 \mu\text{s}$ alignment, which is negligible compared to the 1 ms sample interval at 1 kHz. For implementations without GPS, a phase-angle correction derived from the remote current phasor can be used [5], at the cost of one additional GOOSE exchange.

6 Conclusions and Future Work

The voltage-based pi-section correction (Mode 1) delivers the final α_{50} improvement targeted in TR-13 Future Work 1. The key results are:

- $\alpha_{50}(\text{ag})$: $0.10 \rightarrow 0.07 \rightarrow 0.05 \rightarrow \mathbf{0.03}$ (cumulative 70% reduction from TR-11 baseline).
- $\alpha_{50}(\text{3ph})$: $0.07 \rightarrow 0.05 \rightarrow 0.05 \rightarrow \mathbf{0.02}$ (cumulative 71% reduction).
- Detection threshold: $0.20 \rightarrow 0.12 \rightarrow \mathbf{0.08}$ pu (60% reduction).
- No pre-fault window required: works at fault inception and during dynamic conditions (e.g. busbar transfers).

All improvements are achieved with no changes to the LM estimator or confidence gate; only I_{thresh} and k_{max} are recalibrated.

Future work.

1. **Adaptive threshold** (TR-13 FW#2): replace the fixed 0.08 pu with a function of the estimated restraint current, $I_{\text{thresh}}(t) = k \cdot \hat{I}_{\text{rest}}(t)$, to maintain the security ratio under varying load.
2. **Frequency-adaptive correction** (TR-13 FW#3): extend the Hilbert estimator to jointly estimate frequency and charging amplitude for islanded microgrids where the system frequency may deviate significantly from 50 Hz.
3. **87B/87L coordination** (TR-14 FW#3): validate that the lower 87L threshold does not conflict with the 87B bus differential during busbar transfer (TR-14 dynamic zone reconfiguration), particularly in the T2_TIP topology where the generator is connected to both busbars simultaneously.

References

- [1] PhD Candidate. SAMPB TR-11/2026: 87L line differential — HIF sensitivity floor analysis. Technical report, PhD Thesis Supporting Report, 2026.
- [2] PhD Candidate. SAMPB TR-12/2026: 87L line differential — magnitude override for high-impedance fault sensitivity recovery. Technical report, PhD Thesis Supporting Report, 2026.
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- [4] IEC 60044-7:1999: Instrument transformers — part 7: Electronic voltage transformers, 1999.
- [5] B. Kasztenny, G. Brunello, and J. Harber. A practical guide to the application of line current differential relays. *SEL Journal of Reliable Power*, 1(1), 2010.